Problem Set 5 (two pages)

Statistics 24510-30040 (W25)

Due 9 am Tuesday, February 18.

Requirements

- Upload your pdf named "Lastname-Firstname-hw5.pdf" to Gradescope under Pset5. Be sure to mark the pages corresponding to each problem.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Related references: Rice 12.2-3, Casella and Berger chapter 11.)

1. (*Two-way ANOVA interaction effect*)

   Below is the partial output of an ANOVA table of a Two-way ANOVA model with interaction.

```
Analysis of Variance
Response: Survival
              Df Sum Sq   Mean Sq   F value   Pr(>F)
Poison         1  23.290
Method         1   8.251
Poison:Method  1   0.012
Residuals     44 196.960
```

   (a) Complete the partial ANOVA table. (Note: R command `pf(5.8, df1=3,df2=4)` computes $\mathbb{P}(F \leq 5.8)$ where $F \sim F_{3,4}$.)

   (b) Derive the ANOVA table for the additive (no interaction) two-way ANOVA model (for the same data). Do the p-values differ from the corresponding p-values in the model in(a)? Explain why.

   (c) Which model is better? Is including the interaction more appropriate?

2. (*Two-way versus one-way* ANOVA)

   A baseball league changed the strike zone each year for three years. (Sounds familiar?)
   The data below show the number of home runs (HR) that each of four players hit for the three years.

Y_{ij}	Player 1	Player 2	Player 3	Player 4
Zone 1	23	21	31	15
Zone 2	12	8	25	10
Zone 3	18	15	30	15

   You may input the data to R (partial sample code below) and conduct the analysis below.

   (a) Perform a one-way analysis of variance for the Player effect on HR. Is there a significant Player effect by the analysis?

   (b) Perform a one-way analysis of variance for the Zone effect on HR. Is there a significant Zone effect by the analysis?

   (c) Perform a two-way analysis of variance for the additive effects of Player and Zone on HR. Is there a significant Player effect or Zone effect by the analysis?

   (d) Explain the reasons of getting different conclusions from the above analyses, and which conclusion you prefer.

   (e) Can we use a two-way analysis of variance to test the interaction effect of Player and Zone? Explain.

   Partial sample R commands:

```
> y=c(23,12,18, ... ,15)
> Player=factor(rep(1:4, each=3))
> Zone=factor(rep(1:3,4))
> summary(aov(y~Player))       #  y~Zone, y~Player+Zone, y~Player*Zone
```

3. (*Two-way additive model without replication*)

For the two factor observations (y_{ijj}) in the following table,

Factor A \ B	B-1	B-2	B-3	B-5	B-5
A-1	4.4	4.3	4.0	3.6	3.3
A-2	4.1	4.1	4.0	4.0	3.9
A-3	3.2	3.9	3.8	4.1	4.4
A-4	3.9	4.0	3.8	4.1	4.4

consider the model (with i indexes row factor, j column factor)

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \ \ i.i.d. \sim N(0, \sigma^2), \qquad i = 1, \cdots, 4; \ j = 1. \cdots, 5.$$

(a) Under the constraints $\sum_{i=1}^{4} \alpha_i = 0$, $\sum_{j=1}^{5} \beta_j = 0$ (as in 12.3.1 in Rice), use simple averaging operations (differences, sums and averages) to compute parameter estimates $\hat{\mu}, \hat{\alpha}_i$ and $\hat{\beta}_j$.

(b) Create the 4-by-5 table of residuals $y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j$.

(c) How does the size of the residuals in Part(b) compare to the size of the estimated row and column effects $\hat{\alpha}_i$ and $\hat{\beta}_j$ in Part(a)?

4. (*Sequential* ANOVA)

Below are two sequential ANOVA tables for the same data, obtained from a two factor treatment (A and B) experiment. The data are not shown.

SOURCE	DF	SS	MS
A	3	3.3255	1.1085
B	3	112.95	37.65
A*B	9	0.48787	0.054207
ERROR	14	0.8223	0.058736

SOURCE	DF	SS	MS
B	3	116.25	38.749
A	3		
A*B			
ERROR			

(a) Complete the second table on the right.

(b) What do you conclude about the significance of treatment A effects, treatment B effects, and interactions?

(c) Assuming there are no empty cells, that is, there is at least one observation in every combination of the levels of the two treatments.
What is your guess/hypothesis on the likely pattern (i.e., numbers) of replicates?
Could all treatment combinations have the same number of replicates for this data?

5. (*Additive model and design matrix*)

Consider a study of balanced two-way layout with one replicate: The first factor α has 3 levels and the second factor β has 2 levels, with one observation for each of the 6 possible combinations of the two factors.

(a) Write up the linear model for each observation Y_{ij}, in six linear equations.

(b) The model in (a) can be represented in vector-matrix form $Y = \boldsymbol{X}\theta + \varepsilon$. Write down the full design matrix, the $\boldsymbol{X}$ matrix. clearly defining how the rows and columns are ordered.

(c) What is the rank of your design matrix?

(d) Using the constraints $\alpha_1 = 0, \beta_1 = 0$, obtain the corresponding design matrix $\boldsymbol{X}_1$.

(e) What is the rank of new design matrix $\boldsymbol{X}_1$?

(f) Write up the linear model for each observation Y_{ij} under the constraints $\alpha_1 = 0, \beta_1 = 0$.

(*) (Exercise for fun, do not hand in) What are the estimates for $\hat{\mu}, \hat{\alpha}_2, \hat{\alpha}_3, \hat{\beta}_3$ in (f)?
(Note that these are the common default estimates given by software such as R.)